Using Graphical Models and Multi-attribute Utility Theory for Probabilistic Uncertainty Handling in Large Systems, with Application to the Nuclear Emergency Management

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***Abstract*— Although many decision-making problems involve uncertainty, uncertainty handling within large decision support systems (DSSs) is challenging. One domain where uncertainty handling is critical is emergency response management, in particular nuclear emergency response, where decision making takes place in an uncertain, dynamically changing environment. Assimilation and analysis of data can help to reduce these uncertainties, but it is critical to do this in an efficient and defensible way. After briefly introducing the structure of a typical DSS for nuclear emergencies, the paper sets up a theoretical structure that enables a formal Bayesian decision analysis to be performed for environments like this within a DSS architecture. In such probabilistic DSSs many input conditional probability distributions are provided by different sets of experts overseeing different aspects of the emergency. These probabilities are then used by the decision maker (DM) to find her optimal decision. We demonstrate in this paper that unless due care is taken in such a composite framework, coherence and rationality may be compromised in a sense made explicit below. The technology we describe here builds a framework around which Bayesian data updating can be performed in a modular way, ensuring both coherence and efficiency, and provides sufficient unambiguous information to enable the DM to discover her expected utility maximizing policy.**

I. Introduction

Most decision analyses require numerical inputs the DM is unsure about. This *uncertainty* can derive from physical randomness, which can be modeled using various methodologies, from judgmental estimates of quantities she is unsure about, or from many other sources as noted in [1]. In particular, in the case of an off-site nuclear emergency, uncertainty can derive from, for example, weather forecasts, observation errors in monitoring data, the degree of public agreement about the efficacy with any countermeasure, and the quality of the models used such as those for atmospheric dispersion. The need for uncertainty handling in any nuclear DSS has been recognized for some time [2]. In this work it was also demonstrated that, at least in this context, it is possible to propose an entirely *Bayesian* solution. However in [3] it was recognized that the development of methodologies for uncertainty handling for use in nuclear DSSs that are both formal and practical were still in their infancy. During the intervening years technologies applying Bayesian techniques to this area ([4], [5]) have advanced sufficiently to ensure that if fully formal methods are developed then it will be possible to actually implement those. So it is now timely to revisit this problem. In more recent years some progress has been made in uncertainty handling for nuclear emergency management, for example by including statistical methodologies in components of DSSs that previously were deterministic, as shown in [6]. Also data assimilation using the appropriate statistical methodologies has been included in some analyses, as for the atmospheric dispersion of the radiation [7], with the main aim of reducing uncertainty. However, there still exists no complete formal theory about how this uncertainty should be propagated over the system. In particular, to our knowledge there does not exist a systematical study of the causal link between different statistical models inside a DSS, which can help in recognizing all the uncertainties underlying the system under study. In this paper we describe the development of some new theoretical frameworks which will eventually lead to a defensible as well as feasible methodology for uncertainty handling, that will provide the necessary structure for integrating a network of probabilistic models into a DSS for addressing emergency management in situations like the one illustrating this paper.

In the context of nuclear emergency management the DSS is typically made up of several different components expressing best expert judgment often over very different domains. In such a situation the composite DSS needs to coherently network together judgments about the quantities that are necessary to perform the analysis, provided by different experts overseeing different areas of the problem. For example, an assessment of the safety of a nuclear plant will be based on expert judgments of nuclear physicists and engineers who build their own decision support model. However, the experts that are needed to assess what might happen *after* an accidental release has happened are different. There experts will be atmospheric physicists and local weather forecasters to inform the spread of the contamination, combining their knowledge

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into a different component. Yet another DSS will build on the *effect* this spread of contamination might have, because of the exposure of humans, animals and plants, and because of the introduction of contaminated items into the food chain. At the very end of the chain of consequences doctors and geneticists can be asked to provide information about the *impacts* of the disaster to health on the population as a function of these different exposure patterns: again experts with very different domain knowledge.

From the description above it is therefore apparent that in a DSS many heterogeneous aspects need to be accommodated to include judgments from different experts with heterogeneous backgrounds. Because of the huge dimensionality of this class of problem, it would be highly desirable to obtain a *modular* structure in which every aspect of the overall problem was modeled and subsequently updated independently and in which the relevant experts could provide judgments autonomously. Critical properties needed by such a network would be the abilities to appropriately encode the relevant uncertainty associated with each component, to propagate these uncertainties over the system and so infer the best overall choice which accommodates all quantified uncertainties and structure in a typical user's utility function.

In the following sections we will give conditions when such modularity can be ensured both a priori and after the collection of data. We note that, assuming that the experts provide rational and coherent judgments about their area of jurisdiction, then, by standard decision theory (see [9] for example), the decisions based on these judgments will be rational when the system is modular. In [10] it was noted that after a large scale disaster, for example after a nuclear accident, the lack of integration and collaboration among all the involved people threatens to cause further negative consequences to the event: in particular high levels of anxiety and stress. This kind of threat to rationality can be directly managed through developing the types of protocols we address here, permitting the people involved to be required only to provide quantitative judgments within their own area of expertise and still be confident that their contributes, when combined with others', will provide a coherent picture of the ongoing emergency. They are then able to concentrate on their own element of the problem, to collaborate with experts with the same background and to combine their diverse judgments into a consensual joint approval of the potential risks associated with the features of the problem they oversee.

To achieve this within a Bayesian paradigm, a DSS needs to be able to use agreed collective beliefs about the qualitative relationships in the process, pasting together the outputs of the different modules, each of them having inputs from different sets of experts, into a single probability model. The DM would then be able to use these statements as her own. Then, when certain conditions hold, she would have sufficient input to be able to behave as a single expected utility maximizer. Furthermore she can be confident that the input probabilities she uses are appropriately informed by the judgments of the relevant experts. Once in place she would therefore be able to argue that she has come to her decision drawing upon as fully as possible the data relevant to the decision making process in a way advised by these experts, and so defend her decisions to an external regulator, the general public and politicians.

The paper is structured as follows. Section II, after briefly introducing the architecture of a typical DSS for nuclear emergencies, discusses the need of uncertainty handling in such a system. Section III then defines the notation, reviews the main background material and explains the statistical setting used to model uncertainty. In Section IV an example representing a nuclear DSS is presented to illustrate the conditions necessary to ensure the modularity property needed for such a system. Section V presents a very simple scenario in which the omission of the uncertainty leads to wrong policies. We conclude the paper with a discussion of some ongoing developments in this area.

II. The needs of a nuclear DSS

In the last 25 years, subsequently to the Chernobyl accident, there has been an increasing number of European projects which have aimed to develop a joint DSS to improve the emergency management of any future accidental release of radioactivity. One of the DSSs developed during these collaborations was RODOS, which, as other nuclear support systems, differed from traditional DSSs in the following aspects:

- It was designed to work in several different contexts, depending on where decisions needed to be made;
- It had a client/server architecture, built on a list of modules connected via a communication interface;
- It was designed to explicitly address and solve the issue of uncertainty handling.

Among the other DSSs to support decisions in the case of radioactive contaminations, more recently MOIRA ([11],[12]) and its newest version MOIRA-PLUS ([13]) provide support for the identification of optimal remedial strategies to restore water systems after accidental introduction of radioactive substances.

Before examining how uncertainty can be incorporated into the system, we first present the conceptual architecture underpinning RODOS. As described in [14], this was split into three distinct subsystems:

- Analysing Subsytem (ASY) modules which processed incoming data and produced forecasts;
- Countermeasure Subsystem (CSY) modules which suggested possible countermeasures;
- Evaluating Subsystem (ESY) modules which ranked these countermeasures.

In [3] it was argued that this architecture corresponded well with the Bayesian paradigm for decision-making. The ASY modules build the statistical models underlying the system, the CSY modules identify actions and calculate their potential consequences and the ESY modules compute their expected utilities.

Fig. 1 summarizes this architecture. On the right of the diagram we note two different sets of available actions: one for

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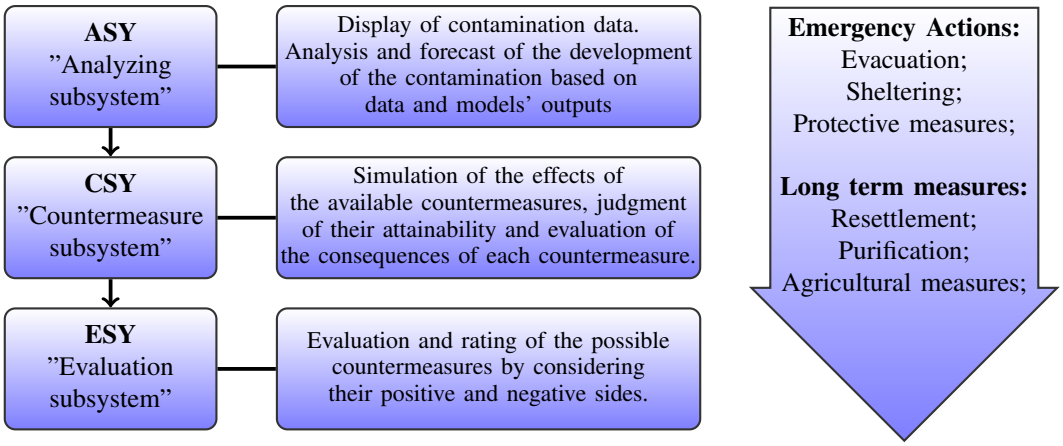

Fig. 1. Conceptual structure of RODOS after [8]

the short term and the other for the long one. Thus RODOS, like certain other nuclear DSSs, provided decision support for all the different phases of a nuclear accident. As time passes the system arrives at different decision points where it must select three modules to form an ASY, CSY and ESY chain appropriate to the context of the decision. In such a DSS it is necessary a temporal control system which manipulates modules in response to the user's requests or to some pre-programmed sequences of desired analyses. These functions were performed by the so called Temporal Support Functions (TSF), providing the flexibility to change operating behavior according to the current phase of the accident, selecting the most suited models and processing and storing incoming data in a suitable way.

Another fundamental element of such a composite DSS is the graphical representation of the forecasts or the estimates deriving from the probability models included in each module, which is handled by the GUI unit. Some of these features are discussed in [15] and references therein. These help the DM having a much clearer idea of how such a complex DSS works, increasing her appreciation and confidence about the results that it provides.

In an integrated DSS like the one described above, it is tempting solely to model expectations of inputs into such a system and treat them as known. However, if there is significant uncertainty associated with such input, then it is well known that such methods can grossly distort the assessment of the comparative efficacy of various competing policies. This will be demonstrated below through a simple example. So to properly inform DMs it is vital to incorporate uncertainty into such a DSS.

Sometimes this uncertainty will be intrinsic to a particular module *within* the panel of domain experts. For example, an estimate of the likely release of radioactivity before it actually occurs is provided by the Source Term Module (STM), which consists of a directed statistical graphical model (see [16] for an introduction) as shown in [3], while one of the available atmospheric dispersion deposition modules was RIMPUFF, a puff diffusion model (see for example [17]). In implemented systems the complexity of the domain demands that such methods are numerical, although occasionally it is possible to perform such uncertainty analytically (see below). We note that the component models may be chosen subsequent to a careful model selection over competing probabilistic models, for example using MAP model selection methods, see e.g. [18]. However, for simplicity we will assume in the following that everyone in a given panel of experts has agreed on a single selected model for each module.

At other times expert judgments are encoded in terms of a collection of sparse sets of simulators over an highly complex deterministic system. This is typical of modules associated with climate change. In the context of systems describing the developing of nuclear incidents such modules might be those which predict the effect of the nuclear accident on the food chain. In these cases uncertainty needs to be assessed by the appropriate panels of analysts using methodologies based on *emulators* which produce such uncertainty measures, analogous to those developed for climate change [19]. In other scenarios there is no model based methodologies possible, so that uncertainty measures need to be elicited *directly* from the panels.

However uncertainties are encoded by the single panels, these uncertainties will need to be communicated to panels using the outputs of that module as the inputs to their own forecasting module. For example uncertainties about the extent of release from a nuclear plant as well as point predictions of point releases will need to be communicated to the panel responsible for the prediction of the contamination spread across the locality. In reality the methodologies behind the calculation of these uncertainties can be very complex. In this paper we will however treat them as a given, centering the analyses on illustrations of what they might look like in very simple situations. This will allow us to concentrate more fully on transparently illustrating how these assessments can be collated, processed and propagated through the network of expert systems.

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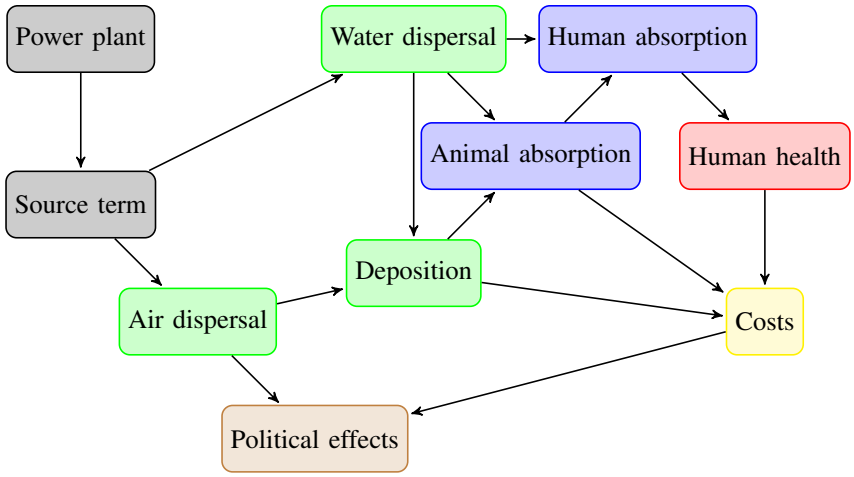

Fig. 2. Plausible network for the modules of a nuclear DSS. The gray vertices are overseen by the engineering panel; the green ones by the environment panel; the blue ones by the biological panel; the brown, red and yellow ones by the political, medical and economical panels respectively

As we will formalize in the following section, the network of panels of experts can often be conveniently depicted through a *directed acyclic graphical model* (DAG), here associated to the common knowledge of all involved people in the development of the DSS. With certain additional assumption this DAG has a fully formal semantic, implying that certain analytical deductions can already be made by the group. A plausible example of such a network that may describe the whole system is presented in Fig. 2, together with a division of the nodes into areas of interest for different sets of experts. Subsequently in this paper we will use this network to illustrate our methodology.

We will also illustrate below the issue of exactly which uncertainties are needed by the composite system to be determined by the purpose of the whole decision support. In particular we argue in this paper that ideally this should be assessed through the structure of the (multi-attribute) *utility function*, which is used to represent and analyze the preferences of the experts and consequently of the DM. There has been an increasing interest in using multi-attribute utility theory (MAUT) in DSSs as noted by the analyses drawing on [20] and [21]: the first studied the contributions of multi-criteria methods (among which MAUT) to DSSs, while the second evaluated the current state of implementation of these methods in DSSs. Both the articles noted the advantages in using such methodologies in the context of supporting decision-making. We here apply these ideas to this complex and necessarily distributed environment.

III. Setting and Notation

We start by introducing the notation necessary for the following. Consider a set of decisions $d \in \mathcal{D}$ available at a certain decision point and let $\mathbf{Y} = (\mathbf{Y}_1, \ldots, \mathbf{Y}_n) \in \mathcal{Y} = \times_{i=1}^n \mathcal{Y}_i$ be a random vector of measurable quantities parametrized by $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ such that $f_{\mathbf{Y}}(\mathbf{y} \,|\, \boldsymbol{\theta})$ is a joint density of the model. Here $\mathbf{Y}_i$, for $i = 1, \ldots, n$, is one of the n modules that are included in the DSS. Let $\{G_1, \ldots, G_m\}$ denote m panels of domain experts, which input the relative information to the DSS necessary for the DM to make the optimal decision. Let also $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n)$ such that $\boldsymbol{\theta}_i \in \boldsymbol{\Theta}_i$ and $\boldsymbol{\Theta} = \times_{i=1}^n \boldsymbol{\Theta}_i$. In a Bayesian framework we assume the parameter vectors $\boldsymbol{\theta}_i$ to be random vectors with density function $f_i(\mathbf{a}_i(d))$, in which $\mathbf{a}_i(d) \in \mathcal{A}_i$ are the hyper-parameters, for $i = 1, \ldots, n$. Note that we are assuming that the random variables of the systems are different for each decision available, with the hyper-parameters being allowed to be functions of d. Each module is under the sole jurisdiction of its associated panel, providing beliefs about the distribution of the appropriate vectors $\mathbf{Y}_i(d)|\mathbf{Z}_i(d)$, where $\mathbf{Y}_i(d)$ and $\mathbf{Z}_i(d)$ are disjoint ($\mathbf{Z}_i(d)$ possibly null) sub-vectors of $\mathbf{Y}(d)$. Depending on the situation, these sets of experts' beliefs may be expressed as conditional densities, as in a full Bayesian analysis, or alternatively, for example, as only some lower order moments of the conditional distributions.

With this notation we can denote the quantities of the network in Fig. 2 as:

- G_1: engineering panel;
- G_2: environment panel;
- G_3: biological panel;
- G_4: political panel;
- G_5: medical panel;
- G_6: economical panel;

- $\mathbf{Y}_1$: power plant;
- $\mathbf{Y}_2$: source term;
- $\mathbf{Y}_3$: air dispersal;
- $\mathbf{Y}_4$: water dispersal;
- $\mathbf{Y}_5$: deposition;
- $\mathbf{Y}_6$: animal absorption;
- $\mathbf{Y}_7$: human absorption;
- $\mathbf{Y}_8$: human health;
- $\mathbf{Y}_9$: political effects;
- $\mathbf{Y}_{10}$: costs.

Panel G_1 (gray) oversees $\mathbf{Y}_1$ and $\mathbf{Y}_2$; panel G_2 (green) oversees $\mathbf{Y}_3$, $\mathbf{Y}_4$ and $\mathbf{Y}_5$; $\mathbf{Y}_6$ and $\mathbf{Y}_7$ are jurisdiction of G_3 (blue); G_5 (red), G_4 (brown) and G_6 (yellow) are responsible for $\mathbf{Y}_8$, $\mathbf{Y}_9$ and $\mathbf{Y}_{10}$, respectively. Assume also that the DM's decision space $\mathcal{D}$ for this example consists of:

- d_1: do nothing;
- d_2: tell the population to shelter;

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- d_3: deliver protective measures;
- d_4: temporarily evacuate;
- d_5: leave the area forever.

Consider also $\mathbf{R} = (\mathbf{R}_1, \ldots, \mathbf{R}_n) \in \mathcal{R}$ a set of attributes such that $\mathbf{R}_i = \mathbf{R}_i(\mathbf{Y}_i(\boldsymbol{\theta}), d)$ (note that also $\boldsymbol{\theta}$ depends on d) and a utility function $\mathrm{U} : \mathcal{R} \times \mathcal{D} \to [0, 1]$: thus, $\mathrm{U} = \mathrm{U}(d, \mathbf{R}_i(\mathbf{Y}_i(\boldsymbol{\theta}_i)), i = 1, \ldots, n)$. Also the utility function and the rewards can depend on the decision d. The attributes $\mathbf{R}_i$'s are introduced for those cases where it is unclear or infeasible to elicit preferences over some $\mathbf{Y}_i$'s. Because our focus is the uncertainty handling and not the optimization problem, in the following we will suppress the dependence on d. The reader should bear in mind though that, depending on the situation, some of the quantities involved may vary across the decisions available.

For the preferences of the DM about the composite system to be expressed in terms of the preferences over different domains of the problem, it is helpful for the composite utility function to factorize respecting the division of the vector $\mathbf{Y}$ into the sub-vectors overseen by the different panels: this concept will become clearer in the following section.

Moreover a system like the one above will usually be dynamic. However, again to limit the technicalities in this short paper we will restrict our discussion to the cross sectional/non-dynamical aspects of the support system. In fact it can be shown that dynamic analogues of the analysis given below can be developed in a straightforward albeit rather technical manner. So the cross sectional analysis we present here forms a template for more nuanced applications. We therefore imagine the DM to be at a particular decision point, for which the vector $\mathbf{Y}$ represents all the relevant modules to the problem. Moreover, at this particular time point, we assume that a dataset $\mathbf{x}$ becomes available, collected from this non-dynamical system. We will discuss at a later point how the ideas we develop in this paper can be generalized into a dynamic framework.

A. The graphical input/output structure

In the previous section we noted that the outputs of some of the modules can be used as inputs for subsequent ones. These relationships between modules can all be depicted through a graphical model where, if the output of module j is used as input for module i, a directed edge from the vertex representing j to the one of i is drawn. If the resulting graph fulfills some constraints then it can be shown that it will be a Bayesian network (BN) (see [22]): now a widely used framework to express beliefs in terms of probabilities. In a Bayesian system this structure forms the framework over which a coherent Analyzing Subsystem can be build. The great success of these graphical models is due to the evocative and intuitive way in which they depict relationships between random variables and to the fast learning and inferential routines that can be performed over them. Let $\mathbf{Z}_i$ be the modules whose output is used as input for the module corresponding to $\mathbf{Y}_i$ and denote $\mathbf{Q}_i = \mathbf{Y}^{i-1} \setminus \mathbf{Z}_i$, where $\mathbf{Y}^{i-1} = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_{i-1}\}$. With this further notation we can define a BN.

Definition 1: A Bayesian network for the random vector $\mathbf{Y} = (\mathbf{Y}_1, \ldots, \mathbf{Y}_n)$ consists of a set of $n-1$ conditional independence statements of the form $\mathbf{Y}_i \perp\!\!\!\perp \mathbf{Q}_i | \mathbf{Z}_i$, together with a directed graph $\mathcal{G}$ having no directed cycles. The directed graph has vertex set $V(\mathcal{G})$ and edge set $E(\mathcal{G})$, where

$$V(\mathcal{G}) = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_n\} \quad \text{and} \quad (\mathbf{Y}_i, \mathbf{Y}_j) \in E(\mathcal{G}) \Longleftrightarrow i \in \mathbf{Z}_j, 1 \leq i < j \leq n.$$

Note the assumption of having a numerically ordered vertex set: that is, $(\mathbf{Y}_i, \mathbf{Y}_j) \in E(\mathcal{G})$ only if $i < j$. Note also that $\mathbf{Y}_i \perp\!\!\!\perp \mathbf{Q}_i | \mathbf{Z}_i$ reads that the vector $\mathbf{Y}_i$ is independent of $\mathbf{Q}_i$ given $\mathbf{Z}_i$: so that the only information to infer $\mathbf{Y}_i$ from $\mathbf{Q}_i$ & $\mathbf{Z}_i$ is from $\mathbf{Z}_i$. Given a directed acyclic graph a BN can be alternatively defined by the recursive factorization formula

$$f_{\mathbf{Y}}(\mathbf{y}) = \prod_{i=1}^{n} f(\mathbf{y}_i \,|\mathbf{Z}_i = \mathbf{z}_i),$$

where $\mathbf{y}_i$ and $\mathbf{z}_i$ are generic realizations of the vectors $\mathbf{Y}_i$ and $\mathbf{Z}_i$ respectively.

Every conditional independence statement of a BN can be deduced from the topology of its graph after a simple transformation, called *moralization*, as stated by the *d-separation* theorem [16]. Recall that a vertex $\mathbf{Y}_i$ is a *parent* of a vertex $\mathbf{Y}_j$, and $\mathbf{Y}_j$ is a *child* of $\mathbf{Y}_i$ in $\mathcal{G}$ if $(\mathbf{Y}_i, \mathbf{Y}_j) \in E(\mathcal{G})$. The set of all parents of a vertex $\mathbf{Y}_i$ is the *parent set*, denoted with $Pa(\mathbf{Y}_i)$, and, for our construction, it corresponds to $\mathbf{Z}_i$. The moralization process simply joins with an edge two vertices in the same parent set that are not connected for an appropriate subset of the graph $\mathcal{G}$. These conditional independences are usually called *Markov conditions* and they state that for every vertex $\mathbf{Y}_i$ in $V(\mathcal{G})$ we have that $\mathbf{Y}_i$ is independent of $\mathbf{Y} \setminus (Dn(\mathbf{Y}_i) \cup Pa(\mathbf{Y}_i))$ given $Pa(\mathbf{Y}_i)$, where $Dn(\mathbf{Y}_i)$ corresponds to the descendant set of $\mathbf{Y}_i$. The descendant set of $\mathbf{Y}_i$ consists of those vertices $\mathbf{Y}_j$ for which there is a directed path on the graph starting from $\mathbf{Y}_i$ and finishing in $\mathbf{Y}_j$.

In Section II we noted that uncertainty can be included into the DSS by introducing non-deterministic relationships between the modules. In the graphical context, one of the simplest such model defines, for example, a multivariate Gaussian distribution over the network, through a set of regression relationships. In the causal literature these are also called *structural equations*. For example in the simplest case when the output variable Y_j is a univariate Gaussian variable then, for $j = 1, \ldots, n$:

$$Y_j = \theta_{0,j} + \sum_i \theta_{i,j} Y_i + W_j, \qquad \forall\, i \in Pa(Y_j), \tag{1}$$

where $\theta_{i,j} \in \mathbb{R}$, for $i < j$, are regression parameter and $W_j \sim \mathcal{N}(0, \psi_j^2)$ independent of the Y_i's and of the other W_i's, for $i < j$. Note that the Gaussianity of the system is exclusively induced by the random errors W_i, $i = 1, \ldots, n$. Thus, different distributions can be envisaged by changing the density function associated to the errors. However, as asserted by the central limit theorem, many situations can be approximately modeled using Gaussian distributions without

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inconsistencies. We note also that the Gaussianity assumption is in no way central to the general development below. It does however provide a class of probability models whose uncertainty handling is particularly straightforward to analyze and so provides us with simple explicit examples demonstrating certain points we need to make.

For Bayesian uncertainty handling each of the parameters $\boldsymbol{\theta}_i$ needs to be treated as a random variable with its own distribution expressing prior knowledge and data provided by the associated expert panel. For the purposes of this paper, we define only some of the moments of the parameters. Let $\mathbb{E}(\theta_{i,j}) = a_{i,j}$ and $\mathbb{V}(\theta_{0,j}) = \tau_{0,j}$ for $i = 0, \ldots, n-1$ and $j = 1, \ldots, n$ with $i < j$. Let also $\mathbb{E}(\psi_i^2) = \lambda_i$ for $i = 1, \ldots, n$. We note that in fact in the current applications of this technology it is usually appropriate to treat these parameters as random processes which change during the progress of the emergency. This generalization is in fact quite straightforward using the class of multi-regression dynamic model [23]: henceforth assume these are fixed in time over the period of the emergency management.

When a graphical model expresses the beliefs of a single expert the common and often default choices are the so called *global* and *local* independence assumptions (see [24],[25]). The global independence assumption states that, if $\boldsymbol{\theta}_i$ and $\boldsymbol{\theta}_j$ are the parameters of $\mathbf{Y}_i$ and $\mathbf{Y}_j$ respectively, then $\boldsymbol{\theta}_i \perp\!\!\!\perp \boldsymbol{\theta}_j$, for all the possible couples i, j. The local independence assumption on the other hand states that if $\boldsymbol{\theta}_i$ is a random vector, then its components are independent of each other, for $i = 1, \ldots, n$. It can be shown that these two assumptions together imply that all the parameters are mutually independent. It is shown in [26] that inferences are usually robust to violations of this assumption if the supporting data sets the panel receives are informative. Various technical issues associated with this independence assumption and its generalizations are discussed in [27].

Under these conditions, the following theorem, from [28], shows how to derive the full distribution of the system, when the variables are defined as in (1).

Theorem 2: Let $\mathbf{Y} = (Y_1, \ldots, Y_n)$ be a random vector, where every Y_i's is defined as in (1). Let Ψ be the diagonal matrix $\Psi = diag(\psi_1^2, \ldots, \psi_n^2)$ and let L be an $n \times n$ upper triangular matrix defined as

$$L_{ij} = \begin{cases} \theta_{i,j}, & \text{if } \{i,j\} \in E(\mathcal{G}), \\ 0, & \text{otherwise.} \end{cases}$$

Set also $B = I - L$, where I is the $n \times n$ identity matrix. Then $\mathbf{Y} \sim \mathcal{N}(B^{-T}\boldsymbol{\theta}_0, \Sigma)$, where $\boldsymbol{\theta}_0 = (\theta_{0,1}, \ldots, \theta_{0,n})$ and $\Sigma = B^{-T}\Psi B^{-1}$.

Since the Gaussian distribution is fully described by its first two moments, an alternative way of stating Theorem 2 is in terms of recursive expressions of the expectations and of the covariances/variances of the vertices. Then the mean relationships are simply obtained by plugging in the expected values for their associated variables. This gives us that

$$\mu_j = \mathbb{E}(Y_j) = \theta_{0j} + \sum_i \theta_{i,j}\mu_i, \qquad \forall\ i \in Pa(Y_j).$$

In order to compute the variances, which will capture the uncertainties about these variables, we first need to derive the covariance relationships between any two random variables. For $k < j$, call

$$\sigma_{k,j} = \mathrm{Cov}(Y_k, Y_j) = \sum_i \theta_{i,j}\sigma_{i,k}, \qquad \forall\ i \in Pa(Y_j). \quad (2)$$

It then follows that the required variances can be calculated using formula (2) as

$$\sigma_{j,j} = \mathbb{V}(Y_j) = \sum_i \sum_k \theta_{i,j}\theta_{k,j}\sigma_{i,k} + \psi_j^2, \quad \forall i, k \in Pa(Y_j).$$

Note that since each parameter $\theta_{i,j}$ in these expressions is itself a random variable, it will also have its own associated uncertainty related, for example, to the quality of the estimate the panel can make of it. However, these other uncertainties can also be folded into these expressions in a straightforward way exemplified later in the paper. Our underlying methodology can be illustrated using the appropriate algebraic relationships like the ones above associated both with the underlying graphical structure and the assumed form of utility functions.

B. Causality and learning

In order to incorporate information deriving from experimental evidence each panel might have available, it has now been recognized that the network like the one described above is not simply a BN, but a causal Bayesian network (CBN) [22]. The relevant concept of causation leading to the CBN was defined in [29]. It has also been recognized that even when the network is not assumed to be causal, a CBN can be derived as a natural extension of a BN in which the global independence assumption is met, as shown in [30]. In a CBN each arc can be interpreted as a direct causal influence between a parent node and a child node, relative to the other nodes in the network: an assumption which has some plausibility within this type of applications (see [22]).

The intrinsic hypothesis behind a causal model is the so-called *manipulation property* [31]. This states that if one of the variables/vectors of the system, $\mathbf{Y}_i$ say, is controlled and set to a particular value, then the outcomes of the remaining variables would occur as if the outcome of the i-th vertex was not controlled. There are often strong arguments for believing that this causal hypothesis will hold in a given scenario. It is this property that allows us at a preliminary stage to conduct experiments, manipulating some of the inputs to each module in a designed experiment and observing the results of these actions, which we assume would be just like what we might observe after a real accident. We note that in practice this assumption of perfect correspondence between designed experiments and observational studies - implicitly a causal hypothesis of the type discussed above - is almost always made in inferences about the inputs of emergency responses in contexts like these.

The databases resulting from these manipulations can be used to start populating the DSS with supporting experimental

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evidence and also to derive the topology of the network connecting the modules. In [32] a Bayesian method to learn CBNs is introduced, which, taking as input a dataset, produce then a causal graphical model based on the data, which can be both observational and experimental. Thus, using this algorithm, a network for the modules can be derived, over which the relative experts provide the required conditional judgments.

Suppose henceforth we now assume that all the users of the DSS agree on the graphical statistical model which connects the modules. This, for example, will be implicit if they accept the network architecture of the composite system.

C. Bayesian updating

The collection of data can reduce the uncertainty in the DSS by the combination of the information in the sample with the prior belief specification of the relative experts. In the full Bayesian framework, this updating is performed through Bayes theorem combining the prior distribution with the sample's likelihood to obtain the *posterior* distribution. All the information in the sample about the parameter of interest, $\boldsymbol{\theta}$ say, is contained in its likelihood function as assured by the *likelihood principle*.

We will now demonstrate how this updating is performed for one of the regressions that define the Gaussian distribution over the graph $\mathcal{G}$ as in (1). The equation for the i-th variable can be rewritten as

$$Y_i = \mathbf{X}_i \boldsymbol{\theta}_i + \varepsilon, \tag{3}$$

where $\mathbf{X}_i$ is a row vector of $k-1$ explanatory variables (we are including a costant term), $\boldsymbol{\theta}_i$ is a column vector of k parameters and $\varepsilon \sim N(0, \psi_i^2)$. Thus, the $k-1$ explanatory variables are the parents in the network and $\boldsymbol{\theta}_i$ contains all the regression parameters. Although this class of models is rather too simple to the typical uncertainty handling needed for nuclear DSSs, it nevertheless provides insight into how new experimental or observational data updates expert judgments concerning the uncertainty of their outputs.

In a full Bayesian framework the experts need to define prior distributions for the parameters. One common choice is to pick *conjugate* distributions, that is ones for which the resulting posterior distribution is in the same family. For a Bayesian Gaussian linear model, as the one we are considering in (3), it is possible to find a conjugate prior, which is the normal-inverse gamma, as we will show. We emphasize that the conjugate analysis below is not essential in later arguments: indeed it will be unusual in our context. However it is invaluable to come to a deeper appreciation of how experimental information feeds through this network of expert systems.

Assume that, after an observational study or a manipulation (we have seen that in the causal framework there is no distinction among the two), a sample consisting of n observations is available and let now $\mathbf{Y}_i$ be a $n \times 1$ column vector, $\mathbf{X}_i$ a $n \times k$ matrix and $\varepsilon \sim N(0, \psi_i^2 I)$. Also recall that in this setting the likelihood is defined to be

$$L(\boldsymbol{\theta}_i, \psi_i^2 | \mathbf{X}_i, \mathbf{Y}_i) \propto \left(\psi_i^2\right)^{-n/2} \exp\left[-\frac{1}{2\psi_i^2}(\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i)'(\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i)\right].$$

We also define the priors. Recall that, if $(\boldsymbol{\theta}_i, \psi_i^2) \sim NIG(\boldsymbol{\mu}_i, \Sigma_i, a_i, b_i)$ (Normal-Inverse Gamma), where $\Sigma_i = \psi_i^2 V_i$ is a positive-semidefinite and symmetric matrix, then its density is

$$f(\boldsymbol{\theta}_i, \psi_i^2) = \frac{1}{2\pi|\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{\theta}_i - \boldsymbol{\mu}_i)' V_i^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\mu}_i)\right) \frac{b_i^{a_i}}{\Gamma(a_i)} (\psi_i^2)^{-(a_i+1)} \exp\left[-\frac{b_i}{\psi_i^2}\right].$$

We now show that this is a conjugate prior for a normal linear model. Using Bayes theorem we have that

$$\begin{aligned} f(\boldsymbol{\theta}_i, \psi_i^2 | \mathbf{Y}_i, \mathbf{X}_i) \propto & \left(\psi_i^2\right)^{-n/2} \exp\left[-\frac{1}{2\psi_i^2}(\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i)'(\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i)\right] \\ & |\Sigma_i|^{-1/2} \exp\left[-\frac{1}{2\psi_i^2}(\boldsymbol{\theta}_i - \boldsymbol{\mu}_i)' V_i^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\mu}_i)\right] \left(\psi_i^2\right)^{-a_i-1} \exp\left[-\frac{b_i}{\psi_i^2}\right] \\ & \propto \left(\psi_i^2\right)^{-n/2-1/2-a_i-1} \exp\left[-\frac{Q}{2\psi_i^2}\right], \end{aligned}$$

where Q is equal to

$$\begin{aligned} Q =& (\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i)'(\mathbf{Y}_i - \mathbf{X}_i\boldsymbol{\theta}_i) \\ & + (\boldsymbol{\theta}_i - \boldsymbol{\mu}_i)' V_i^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\mu}_i) + 2b_i \\ =& \boldsymbol{\theta}_i'(\mathbf{X}_i'\mathbf{X}_i + V_i^{-1})\boldsymbol{\theta}_i + \boldsymbol{\theta}_i'(\mathbf{X}_i'\mathbf{Y}_i + V_i^{-1}\boldsymbol{\mu}_i) \\ & + (\mathbf{Y}_i'\mathbf{X}_i + \boldsymbol{\mu}_i' V_i^{-1})\boldsymbol{\theta}_i + (\mathbf{Y}_i'\mathbf{Y}_i + \boldsymbol{\mu}_i' V_i^{-1}\boldsymbol{\mu}_i + 2b_i) \\ =& (\boldsymbol{\theta}_i - \boldsymbol{\mu}_i^*)'(V_i^*)^{-1}(\boldsymbol{\theta}_i - \boldsymbol{\mu}_i^*) + 2b_i^{**}. \end{aligned}$$

Note that we have defined

$$\begin{aligned} V_i^* &= (V_i^{-1} + \mathbf{X}_i'\mathbf{X}_i)^{-1}, \\ \boldsymbol{\mu}_i^* &= V_i^*(V_i^{-1}\boldsymbol{\mu}_i + \mathbf{X}_i'\mathbf{Y}_i), \\ b^{**} &= \mathbf{Y}_i'\mathbf{Y}_i + \boldsymbol{\mu}_i' V_i^{-1}\boldsymbol{\mu}_i + 2b_i - (\boldsymbol{\mu}_i^*)'(V_i^*)^{-1}\boldsymbol{\mu}_i^*. \end{aligned}$$

We can consider again the posterior distribution to derive that

$$\begin{aligned} f(\boldsymbol{\theta}_i, \psi_i^2 | \mathbf{X}_i, \mathbf{Y}_i) \propto & \left(\psi_i^2\right)^{-n/2-a_i-1} \exp\left[-\frac{b_i^{**}/2}{\psi_i^2}\right] \\ & \times \left(\psi_i^2\right)^{-1/2} \exp\left[-\frac{1}{2\psi_i^2}((\boldsymbol{\theta}_i - \boldsymbol{\mu}_i^*)' (V_i^*)^{-1}(\boldsymbol{\theta}_i - \boldsymbol{\mu}_i^*))\right] \\ \sim & NIG(\boldsymbol{\mu}_i^*, \Sigma_i^*, a_i^*, b_i^*), \end{aligned}$$

where $a_i^* = a_i + n/2$, $\Sigma_i^* = \psi_i^2 V_i^*$ and $b_i^* = b_i^{**}/2$.

Through this kind of updating the sample information can be introduced into the DSS for every module and the

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overall uncertainty reduced, leading to a more reliable decision making. Note that as more information is input into the system, Σ_i^* becomes systematically smaller. Even when no such conjugate analysis is possible, current sampling methods allow us to calculate posterior distributions numerically from which various posterior moments we use later in this paper can be drawn. Within this context these numerical distributions, as their moments, can be delivered directly into the DSS. The properties described above, where uncertainties reduce as data increases, are generic to a wide range of Bayesian models whether or not they are Gaussian.

We note further that when the quantities of interest are only some of the moments but the system is non-Gaussian it is then possible to use techniques of Bayesian linear estimation (see [33]) to provide coherent and simple algebraic inputs to the system. However, because of the space constraint and the difficulty of illustrating such procedures, these kind of updating will not be included here. The effect of relevant experimental evidences with the necessary causal assumptions will then reduce the variance of output variables in a well defined quantifiable way, which, again, will be used as inputs to other modules in the system.

D. The combination of beliefs

One of the main issues for maintaining the integrity of large DSSs is the fact that many different sets of people introduce information into the system. There exists a quite large literature on how to combine beliefs provided from different people into a single coherent statement. See [34] for a recent review. However, it mostly focuses on the case in which everyone provides information about the whole system, which is a different setting from the one considered in this paper. We now however review the main results in the area, since in the next section we will show that with the technology we are developing, some of the properties can hold also in this more composite case.

When requiring that the DM acts as if she is a single expected utility maximizer who accommodates all information provided by the involved experts it is necessary for the DSS to be externally Bayesian [35]. Loosely speaking external Bayesianity (EB) demands that an aggregated posterior distribution is the same either if every expert updates via Bayes Formula his prior distribution and then the aggregation is performed and if the prior distributions are combined and then Bayes theorem is applied to obtain a single posterior distribution. We have argued above that this property is critical if the decision of the DM are going to be defensible to an outside auditor.

In [36] it is shown that under a certain set of conditions the only non-dictatorship EB pooling operator, which performs the aggregation of a set of densities, is the logarithmic operator (LogOp). If the i-th member of a pool of k members provides a probability distribution $f_i(\boldsymbol{\theta})$, then the LogOp is defined as

$$\bar{f}(\boldsymbol{\theta}) \propto \prod_{i=1}^{k} f_i^{w_i}(\theta)$$

where $w_i \geq 0$, for $i = 1, \ldots, m$, are arbitrary constants such that $\sum_{i=1}^{m} w_i = 1$. The constant w_i is a weight representing the level of trust of the expert that delivered the density f_i.

In [27] a new class of logarithmic operators were introduced, which possesses a generalized version of EB, called conditional external Bayesianity (CEB). Heuristically speaking, CEB requires the same indifference about the order of aggregation and prior-to-posterior analysis for the conditional distributions associated to a BN $\mathcal{G}$. This can be obtained if the likelihood factorizes in the same way as the density function associated to $\mathcal{G}$. A generalization of the LogOp has now been developed which, instead of pooling marginal distributions, pools conditional ones. Without considering the details related to measure theory, this generalization is quite straightforward and leads to the *conditional LogOp* defined as

$$T_j(f_{1j}, \ldots, f_{mj})(\mathbf{y}_j \,|A_j) \propto \prod_{i=1}^{m} [f_{ij}(\mathbf{y}_j \,|A_j)]^{w_{ij}(A_j)},$$

where w_{ij} are the weights, possibly function of A_j, adding up to one, $A_j = \{Pa(\mathbf{Y}_j) = Pa(\mathbf{y}_j)\}$ and f_{ij} is the conditional distribution relative to $\mathbf{Y}_j$ provided by the i-th expert. Recall that in the framework of DSSs most of this quantities can be a function of the decisions d, and consequently be elicited for each decision available.

Two other works are worth noting here. In [37] the use of graphical models to aid the process of aggregating beliefs is investigated. In particular, it was proved there that a pooled distribution obtained with the LogOp reflects any shared Markov independence by the group and for this reason, it can be represented in a concise and natural manner with a BN as well. However, it was also noted that any pooling operator cannot maintain all independences representable in a BN. On the other hand in [38] a new framework of opinions' aggregation for linear Bayes theory was developed, based on expectations rather then densities. After redefining variations of the main properties present in the literature about the aggregation of densities, it was shown there that an expectation aggregated through a linear combination of expectations is coherent if each of the experts has given a coherent specification. These combination methods are therefore now sufficiently well developed to integrate into the DSS we envisage in this paper.

E. Multiattribute utility theory

Multi-attribute utility theory provides independence concepts for preferences over a large number of attributes. Again in any Bayesian DSS the typical form of the user's utility function will provide the framework from which the Evaluation Subsystem can be quantified.

When no preferential independences hold, it is difficult for a DM to express her preferences coherently. On the other hand, the recognition of independences reduces the dimensionality of the space over which she has to express preferential judgments, simplifying considerably the elicitation of her utility function. For simplicity here we assume that $\mathbf{R}_i \equiv \mathbf{Y}_i$, for $i = 1, \ldots, n$. We also suppress the dependence on d. Denote also with $\boldsymbol{Y}_A$ the subset of the component attributes whose indices lie in a

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subset $A \subseteq [n] = \{1, \ldots, n\}$ and let also $\bar{A} = [n] \setminus A$ denote the complement in $[n]$ of the set A.

In utility theory there are many different definitions of independence and we define them here through the factorization of the utility function that they induce (for more references see [39]).

Definition 3: The set of components attributes $\mathbf{Y}_A$ is said to be *utility independent* of the other attributes $\mathbf{Y}_{\bar{A}}$ ($\mathrm{UI}(\mathbf{Y}_A, \mathbf{Y}_{\bar{A}})$) if the joint utility function can be factorized as

$$\mathrm{U}(\mathbf{y}) = f(\mathbf{y}_{\bar{A}}) + g(\mathbf{y}_{\bar{A}})\,\mathrm{U}(\mathbf{y}_A), \qquad g(\cdot) > 0,$$

where f and g depend only on their arguments and $U(\mathbf{y}_A)$ is a utility function not depending on the attributes $\mathbf{y}_{\bar{A}}$, which are fixed to a reference value.
The utility independence condition $\mathrm{UI}(\mathbf{Y}_A, \mathbf{Y}_{\bar{A}})$ demands that the preferences over the attributes $\mathbf{Y}_A$ are the same for every level at which the attributes $\mathbf{Y}_{\bar{A}}$ can be fixed.

When UI can be verified for every attribute in the system or for every subset of the attributes' set, then the joint utility function simplifies remarkably.

Theorem 4: Let $2^{[n]}$ be the set of all subsets of $[n]$. Given attributes $\mathbf{Y} = (\mathbf{Y}_1, \ldots, \mathbf{Y}_n)$, with $n \geq 2$ and values $\mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_n)$, a *multi-linear* utility function

$$\mathrm{U}(\mathbf{y}) = \sum_{A \in 2^{[n]}} k_A \prod_{a \in A} \mathrm{U}(\mathbf{y}_a),$$

exists if and only if $\mathbf{Y}_a$, $a = 1, \ldots, n$, is utility independent of its complement $\mathbf{Y}_{\bar{a}}$, where the utilities U are functions of their arguments only and the k's are the *criterion weights*.
The criterion weights represent the importance of the terms in the previous utility factorization (for a definition see [22] and [9]). For example, if a high value is assigned to k_i, it follows that a value of this attribute with a high utility is desirable. Note that, calling $\mathbf{y}^0$ and $\mathbf{y}^*$ the worst and the best possible values of $\mathbf{y}$ respectively, if one assigns $\mathrm{U}(\mathbf{y}^0) = 0$ and $\mathrm{U}(\mathbf{y}^*) = 1$, then all the criterion weights are in $(0, 1]$.

In the multi-linear factorization multiplicative terms are included: these are able to represent the interactions among some of the attributes. If these interaction terms are not considered relevant, so that the overall utility depends only on the marginal values, then the attributes verify a different class of independence, called *additive independence*. This assumption is the most used in practice because it is simpler to explain and enact.

Definition 5: Let $\mathbf{Y}_1, \ldots, \mathbf{Y}_r$ be disjoint subsets of $\mathbf{Y}$, such that $\cup_{i=1}^{r} \mathbf{Y}_i = \mathbf{Y}$. We say that $\mathbf{Y}_1, \ldots, \mathbf{Y}_r$ are additive independent (AI) if the joint utility function can be written as

$$U(\mathbf{Y}) = \sum_{i=1}^{r} f_i(\mathbf{Y}_i).$$

IV. Modularity over the network

Expected utility theory tells us that the optimal decision for a DM is the one maximizing her expected utility. We note that this expected utility can often be expressed as a function of various moments of the random variables. For example, in the linear case for which $U(y_i) = y_i$, the expected utility of Y_i corresponds to the expectation of Y_i. In the following section we will show some results relating the shape of the utility function and the moments necessary to derive its expectation. Moments are functions of the parameters of these random variables and in the multi-attribute case we need conditions on the interactions between these parameters if we want to obtain a modular structure for the DSS. In particular we now require that global and local independence assumptions hold and that the preferences respect an additive factorization. We again refer here the reader to [27] for a discussion of the local and global independence assumptions in this context.

This kind of assumptions is usually discussed and, eventually, agreed during *Facilitated Workshops* and *Decision Conferences* (see [40]), in which representatives of different stakeholders, appropriate scientists and authorities involved meet. The prime purpose of these meetings is the elicitation of an agreed utility function between potential users in an emergency. This is elicited using some simple examples of hypothetical scenarios to get the group's preferences appropriately calibrated. Once this utility function, and in particular the criterion weights, are fixed, the different groups of experts individually continue to draw out their beliefs about the process, as described in the previous sections.

For the final developments assure that for simplicity each panel consists of one member only. When these assumptions hold, the relevant attributes are informed by a particular panel that will ideally provide all involved in the decision making with the posterior probability distributions or summaries thereof associated with the outputs of their particular domain. Within the context of the regression models these assessment would include posterior distributions on the regression parameters or, more generally, on the parameters of the conditional distributions of the attributes under their jurisdiction.

Fig. 3 shows the expected utility scores of the different decisions available to the DM for the nuclear example illustrated with the network in Fig. 2, under the conditions just introduced. Decisions d_3 (protective measure) and d_4 (evacuate) obtained the highest scores, with a significant distance from the other decisions available. In this situation the overall expected utility of a decision is a function of the panels' expected utilities and consequently one can analyze the scores associated with the different panel domains for the single decisions.

Fig. 4 shows the panels' expected utility scores for the decisions of evacuating and delivering protective measures. Recall that the overall expected utility obtained by a decision is a weighted average of the relative scores shown in this graph. We can note that providing protective measures obtained a way higher score in the economical area comparing it to the evacuation option, because of the high costs consequently to transfer a high number of people. However, the decision of evacuating performed far better in the biological and the medical modules, because of the smaller intake of radiations for the population. On the other modules, the results are the

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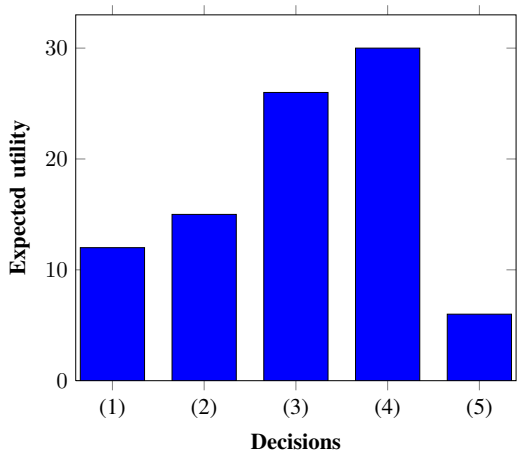

Fig. 3. *A priori expected utility scores*

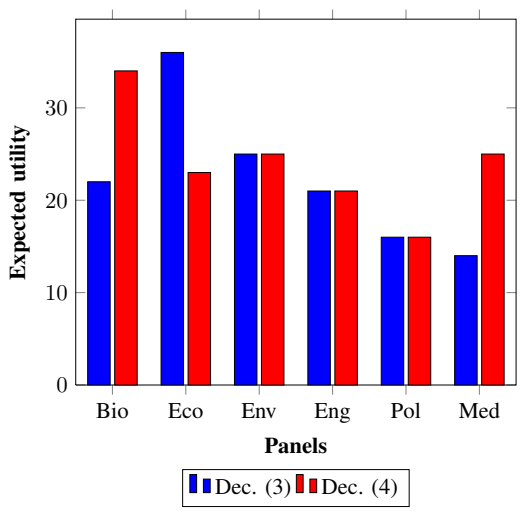

Fig. 4. A priori panel scores for the two best decisions

same for the two options and from these considerations (we assume the weights to be the same for the different areas) follows the fact that d_4 obtained an higher overall expected utility score.

The type of analysis illustrated through this simple example relies on various assumptions. Prior to an actual accident different panels will bring the posterior distributions associated with their domain of expertise. These can also be updated in the light of further experiments before the actual emergency starts. When an emergency occurs, these judgments are fed into the DSS and combined with the utility function elicited during the Decision Conferences to help elucidating *real* scenarios faced by the group's directly countermeasures. We have assumed that these judgments about parameters are mutually independent of each other: otherwise any formula associated with uncertainty handling may no longer be valid. There are two simple conditions that can be imposed on the system that will allow this assumption. The first is that any experimental data used by the different panels is associated with experiments that are mutually independent of each other. A second kind of legitimate data set is one which is observational but complete. Both these scenarios lead to a likelihood which separates over the different panels' parameter vectors. This assumption will then continue to be valid as the actual incident proceeds, provided that information about the progress of the incident is complete until that point.

There are two other assumptions. The demand that there is only one expert for each panel can be initially generalized if the various panel members can agree to deliver their opinions in terms of a *single* probability distribution. In practice this may be difficult to achieve or even inappropriate to the context, although in an earlier section we reported results that demanded it if the coherence of the system was going to be ensured. Furthermore recent developments in decision support theory have made such delivery at least plausible. In particular, the elicitation of the quantities necessary to compute the scores of the individual panels can be performed during other Facilitated Workshops, in which only the experts concerning that particular module meet and discuss the features of the models available. We note that the recent explosion of available Bayesian software has made this argument much more achievable, because scientists are now much more familiar with such probabilistic outputs.

Lastly, if we move our discussion from the additive case to the multi-linear case for utility functions, we add interactions between the preferences of different panels. Sadly, often to be realistic it is necessary to consider this situation. In this case the expected utility might no longer lead to the useful factorization of the expected utility we employed in this example. Fortunately in the next section we will show the polynomial associated to a simplification of the running example for a multi-linear factorization and the issues that may arise from the misuse of uncertainty in this scenario.

V. Consequences of the omission of uncertainty

Consider a simplification of the example in Fig. 2, in which only the human health and costs modules are included. Assume also that each of the two modules consists of a single random variable. The variable for human health is Y_1, say, while the one for costs is Y_2: they can correspond for example to the number of people experiencing negative symptoms and the financial implications to the government, respectively. Their relationship is described by the network in Fig. 5: in this case the graph includes the relationships among the parameters that will appear in the expected utility function when the variables are define as in (1).

Assume that $\mathrm{U}(y_1, y_2) = y_1 y_2$ in this case (a multilinear factorization without the additive component) and that the decisions available are either to evacuate (d_1) or to provide protective measures (d_2). It can be shown that in this case the expected utility is of the form

$$a_{0,2}a_{0,1} + a_{1,2}[\lambda_1 + \tau_{0,1} + a_{0,1}^2], \tag{4}$$

where all the quantities in the previous expression depend on d. The optimal decision will be then the one minimizing

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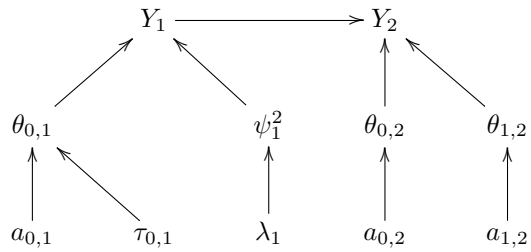

Fig. 5. Network for the simplified example of a nuclear DSS

expression (4) (since the attributes represent negative consequences, the optimal decision is the one minimizing the expected utility). Now assume that the values provided for the decision of evacuation are:

$$a_{0,1}(d_1) = 2, \quad a_{0,2}(d_1) = 2.5, \quad a_{1,2}(d_1) = 2,$$
$$\lambda_1(d_1) = 4, \quad \tau_{0,1}(d_1) = 4,$$

while for the protective measures option we have that:

$$a_{0,1}(d_2) = 2.5, \quad a_{0,2}(d_2) = 2, \quad a_{1,2}(d_2) = 2,$$
$$\lambda_1(d_2) = 2, \quad \tau_{0,1}(d_2) = 1.$$

For the purposes of this simple example, which aims to show the consequences of omitting the uncertainty of the estimates, we will not focus on the heterogeneity of the experts that provide these estimates and we will simply assume that they are given by a single person. The hyper-parameter $a_{0,1}$ corresponds to the expectation of the expected number of people with adverse symptoms, while $a_{0,2}$ represents the expectation of expected amount of financial cost. It is then reasonable to provide a higher value for $a_{0,1}$ in the case of protective measures than in the evacuation scenario. By the same logic, even though protective measures may lead to more expenses for medical cures, the costs of an evacuation are incredibly high and significantly greater than for the other available option. The hyper-parameter $a_{1,2}$ represents the strength of the causal relationship between the two variables. A priori, the expert(s) may not be very well informed about this quantity and decide to provide the same value for the two decisions available. The expectation of the variance of the number of people with negative symptoms is λ_1: in the case of delivering protective measures this value is definitively lower, since it is easier to provide shelter to most of the population. On the other hand during an evacuation, it may not be possible to help all the people in danger and only a subset of them may be moved to safer areas: as a consequence, the expectation of the variance of Y_1 for decision d_1 is higher. The last hyper-parameter $\tau_{0,1}$ corresponds to the variance of the expectation of Y_1 and, by the same argument, it is higher in the evacuation case than in the protective measures one.

In Section II we noted that, once the expectations are provided, then it is possible to treat them as known quantities, so that the analysis becomes very similar to a deterministic one. In this example the hyper-parameters expressing expectations are $a_{0,1}$, $a_{0,2}$ and $a_{1,2}$, while $\tau_{0,1}$ and λ_1 refer to variances. To show the issues deriving from the suppression of the uncertainty of the estimates, imagine that the values for $\tau_{0,1}$ and λ_1 are set to zero, so that the expression to minimize is now

$$a_{0,1}a_{0,2} + a_{1,2}a_{0,1}^2.$$

In this case, plugging in the values provided by the appropriate expert(s), the expected utility for the evacuation scenario is 13, while the expected value for delivering protective measures is 17.5: the optimal choice, when no variation in the estimates is allowed is to evacuate the population. Consider now the actual expected utility expression as in (4), including the second-order uncertainty. In this situation the expected utility of d_1 is 29, while the expected utility under d_2 is 23.5, from which it follows that the optimal decision consists of delivering protective measures. The misjudgment of uncertainty in this case has led to an indefensible decision, highlighting the need to handle uncertainty at every level.

VI. Discussion

The implementation of Bayesian decision theoretic techniques into DSSs has been in many cases challenging and often considered too complex to be actually developed. In this paper we have shown that there are some conditions where the non-deterministic framework do not bring too many issues in the modeling of decision-making problems. In these cases software, ad hoc or already implemented methods, can be used to introduce uncertainty in the single modules. There are also several pieces of available software which encodes statistical graphical modeling that can be embedded into the DSS to describe the relationships among the modules. Moreover, we have shown through simple examples the threats to rationality that can arise from an omission of the uncertainty component in a decisional process. We note that, although uncertainty handling concerning most nuclear DSSs has vastly improved after recent years [41], the formal incorporation of these methods within the composite whole is still in its infancy. The methodological developments illustrated in this paper are now being used to structure protocols for Decision Conferencing of DSSs improvements in this important area.

Due to the space constraint we have had to restrict ourselves to the discussion of non dynamical systems at a single decision point. However, the technologies we are developing do not require such non-dynamic systems to be valid. In fact a technology that generalizes the apparatus discussed here and incorporates these necessary features is straightforward to specify and will reported in a later paper. For example, the multi-regression dynamical model [23] guarantees under a certain set of conditions the modularity of the probability distribution over the network at every time point. Moreover, the study of decisional processes, such as the Markov decision process, can also be incorporated into the modeling of several time decision points.

To aid the efficient application of more general techniques, the authors are also currently investigating the use of computer algebra to simplify the polynomials resulting from the

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expected utilities when the relationships through the modules are defined as in (1). In the last few years computer algebra and algebraic geometry have been used with success in different areas of statistics and we believe that such a technology can solve many of the problems faced in DSSs. Indeed, the expression resulting from the expected utility computations are often intractable and a higher level of abstraction seems to be necessary in order to be able to identify the relevant quantities for the individuation of an optimal policy.

Finally, we note that probabilistic DSSs that use the technologies we employ here are not restricted to the management of nuclear emergency response. The authors are also currently applying their methods to network of expert judgments associated with food security. Again details of this new application will be reported in a future paper.

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